Lecture - 3

❖ Stages of Problem Solving

The six phases of problem-solving in C programming form a structured framework called the Program Development Cycle, which includes **Problem Definition, Problem Analysis, Algorithm Design, Coding, Testing/Debugging,** and **Documentation/Maintenance.**

1. Problem Definition

- ➢ This is the first step, where we clearly identify what problem needs to be solved.
  - ✓ Understand the main objective.
  - ✓ Define exactly what the program should accomplish.
  - ✓ Avoid misunderstanding the problem.
  - ✓ Set clear and specific goals.
- ➢ *Example:* Develop a program to calculate the area of a circle.

2. Problem Analysis

- ➢ In this stage, we determine what information is needed and what result should be produced.
  - ✓ Identify the inputs.
  - ✓ Identify the required outputs.
  - ✓ Determine data types such as int, float, or char.
  - ✓ Identify the mathematical formulas or rules required.
- ➢ *Example:*
  Input: Radius (float)
  Formula: Area = $\pi \times r^2$
  Output: Area (float)

3. Design – Algorithm & Flowchart

- ➢ Here, the solution is planned before writing the actual code.
  - ✓ Develop a step-by-step algorithm.
  - ✓ Write pseudocode in simple English.
  - ✓ Draw a flowchart to visually represent the process.
  - ✓ Divide a complex problem into smaller modules.
  - ✓ Consider efficiency in terms of time and memory.
- ➢ *Example:*
  Start → Read radius → Calculate area → Display area → Stop.

4. Implementation – Coding

- ➢ In this stage, the designed solution is converted into an actual C program.
  - ✓ Open a text/code editor.
  - ✓ Convert the algorithm or pseudocode into C syntax.
  - ✓ Use appropriate variables, functions, operators, and statements.
  - ✓ Add meaningful comments to make the code understandable.

- ✓ Save the program with a .c extension.
- ➢ *Example:* circle_area.c

5. **Testing and Debugging**
   - ➢ After coding, the program must be compiled and tested.
     - ✓ Compile the C program.
     - ✓ Correct syntax errors.
     - ✓ Resolve header file and linking errors.
     - ✓ Test using different inputs.
     - ✓ Identify and fix logical and runtime errors.
   - ➢ *Example:* Test the circle program with radius values such as 5, 10, 0, and negative values to check how the program behaves.

6. **Documentation and Maintenance**
   - ➢ This is the stage where the program is explained, maintained, and improved after development.
     - ✓ Prepare user documentation or manuals.
     - ✓ Document important functions and code structures.
     - ✓ Fix bugs discovered later.
     - ✓ Modify or update the program when requirements change.
     - ✓ Improve performance and usability when necessary.

Sample Important Questions:

1. What are syntax errors?
2. What is debugging?
3. What are logical errors and runtime errors?
4. Explain the six stages of the Program Development Cycle with a suitable example.
5. Explain Problem Definition, Problem Analysis, and Design stages in detail.
6. Explain the importance of Algorithm, Pseudocode, and Flowchart in problem solving.
7. Explain Testing and Debugging, including syntax, logical, and runtime errors.
8. Explain Documentation and Maintenance and their importance in software development.
9. Develop the problem definition, problem analysis, algorithm, and flowchart for finding the area of a circle.
10. Prepare the complete Program Development Cycle for calculating the average of three numbers.
11. Identify the inputs, outputs, data types, and formula required to calculate the area of a circle.
12. Write an algorithm and flowchart for calculating the simple interest.
13. Explain how you would test and debug a program using different input values.